Varad Pendkar

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Professional Summary

AI & ML undergraduate with hands-on experience building and deploying machine learning systems across NLP, computer vision, and generative AI. Projects span RAG pipelines, real-time object detection, ensemble classification, and LLM-powered automation — with deployments using Flask, Streamlit, and Gradio. Graduating May 2026, targeting roles in ML Engineering, Data Science, or AI development.

Technical Skills

Programming & Tools: Python, SQL, HTML/CSS, Git, GitHub
Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV
Machine Learning: Supervised & Unsupervised Learning, Ensemble Methods, Hyperparameter Tuning, Cross-Validation, Transfer Learning
NLP & Generative AI: Tokenization, Embeddings, Transformers, Prompt Engineering, LLM Integration, RAG, LangChain, Groq, ChromaDB
Computer Vision: Object Detection, Image Classification, Real-Time Inference, MediaPipe, YOLO, OpenCV Pipelines
Deployment & MLOps: Flask, Streamlit, Gradio, GitHub Actions, MLflow, Docker

Projects

AI-MEDPAY — Intelligent Insurance Recommendation System — GitHub

Tech: Python, Flask, LightGBM, Pandas, NumPy, OCR, SQLAlchemy, Jinja2, Tailwind CSS

- Built an AI-powered insurance recommendation platform ranking personalized health insurance plans using structured user inputs and ML scoring.
- Implemented LightGBM-based ranking model for plan suitability and integrated OCR-based medical bill extraction for cost insight generation.
- Developed a responsive full-stack web application using Flask, Jinja2, Tailwind CSS, and REST APIs with secure backend architecture.

Automated EDA Platform (LLM-powered) — GitHub

Tech: Python, Pandas, Seaborn, Gradio, Mistral LLM

- Developed an automated EDA system converting raw datasets into **natural-language insights and visual analytics** using Mistral LLM.
- Reduced manual EDA effort by **70%** via LLM-driven cleaning, imputation, encoding, correlation analysis, and statistical summaries.
- Deployed an interactive **Gradio-based UI** for dataset upload and instant report generation with error handling.

RAG Document Chatbot — LangChain + Groq + ChromaDB — GitHub

Tech: Python, LangChain, Groq, ChromaDB, HuggingFace

- Built a RAG pipeline answering natural language queries from custom PDFs using LangChain and Groq's LLaMA 3.3 70B.
- Implemented document ingestion, text chunking, and semantic embedding with HuggingFace models in a persistent ChromaDB vector store.
- Integrated RetrievalQA chain for context-relevant chunk retrieval and grounded response generation with source traceability.

House Price Prediction — Multi-Algorithm Regression Pipeline — GitHub

Tech: Python, Scikit-learn, Flask, HTML/CSS, Joblib

- Built a regression pipeline comparing 13 algorithms including LightGBM, XGBoost, and ANN Regressor with sklearn Pipeline for leakage prevention.
- Deployed a full-stack Flask + HTML/CSS web application with Joblib model serialization for live predictions.

Real-Time 3D Object Detection from Video — GitHub

Tech: Python, MediaPipe, OpenCV, TensorFlow

- Built a real-time pipeline detecting and tracking 3D objects (shoes, mugs, cameras) using MediaPipe Objectron with frame-level inference.
- Implemented bounding-box tracking and landmark detection via Holistic and Hand models, optimized for low-latency continuous video streams.

Education

Bachelor of Technology — Computer Science Engineering (AI & ML) *Aug 2022 – May 2026*
G. H. Raisoni College of Engineering and Management, Pune | CGPA: 7.23

Certifications

Foundations: Data, Data, Everywhere — Google (Coursera) | Covered data types, data lifecycle, SQL basics, and analytical thinking fundamentals.